Shyam Kumar Reddy Kalla

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Experience

AltiusHub Nov 2024 – Present

Software Development Engineer I

- Architected a microservice track-and-trace compliance platform from scratch across **6 markets** (Saudi Arabia, Indonesia, Azerbaijan, Egypt, UAE, Bahrain; 4 live, 2 in progress), automating 26 regulatory report types per country, one country per sprint for 4 enterprise customers.
- Executed zero-downtime migration to declarative list-based partitioning on 50M+ row PostgreSQL tables via logical replication, cutting p99 latency from **1.2s to under 500ms** (∼60%) and improving vacuum performance by ∼40%.
- Integrated the platform with Microsoft Dynamics 365 ERP, syncing **5K records/day** and saving ∼2 hrs/day of manual work for 3 enterprise clients.
- Offloaded long-running processing from the request-response cycle to Celery and RabbitMQ background workers (retries, distributed execution), improving processing throughput by **35%**.
- Created Smart Migrate, a Django management command that fixes migration drift (migration histories diverging across environments) in GitHub Actions deployments across 21 edge cases, cutting migration time from **2 hrs to 15 mins**.

Rockfort Labs Jul 2023 – Dec 2023

Machine Learning Intern

- Built anomaly-detection models for industrial-boiler predictive maintenance on real-time IIoT sensor streams, benchmarking 8 approaches on precision/recall and reaching **98%** accuracy per boiler on held-out data.

Projects

Pulse | **Live Demo** | Python, Django/DRF, Celery, Redis, PostgreSQL, Next.js, Docker, Kubernetes

- Designed a multi-tenant, multi-channel notification platform (Email/SMS/Push) with state-driven delivery, priority routing, exponential-backoff retries, idempotency keys, Redis rate limiting (10 req/min per user/channel) and dead-letter queues; **50+ tests** across 24 failure and retry scenarios.
- Deployed independently scalable Django API and Celery worker services on Docker and Kubernetes HPA (**2–10 replicas** at a 50% CPU target), validated across 7 workers, plus a Next.js operator console (delivery logs, routing, metrics).
- Load-tested with Locust, sustaining **60+ RPS** at 50 concurrent users (p95 under 150ms, p99 340ms) and keeping errors below 1% under 200+ concurrent users in 10-minute stress tests.

SentinelPay | Java 21, Spring Boot, PostgreSQL, Redis, Kafka, Docker, Prometheus, Grafana

- Engineered an ACID-compliant double-entry ledger with immutable journal entries and integer monetary precision, preventing double-spending via deterministic row-lock ordering, `SELECT FOR UPDATE` and database constraints.
- Implemented idempotent transfers with Redis atomic locks and PostgreSQL unique constraints, plus a transactional outbox (`FOR UPDATE SKIP LOCKED`) and Kafka consumer deduplication for reliable publishing and recovery.
- Developed an end-of-day reconciliation engine using streaming CSV/NDJSON parsing and bounded Java 21 virtual-thread concurrency, processing **100K records** and detecting 100 anomalies in 3.084 seconds; 35 integration tests on PostgreSQL, Redis and Kafka.

Technical Skills

Languages: Python, Java, TypeScript/JavaScript, SQL
Backend & Frontend: Django, Django REST Framework, Spring Boot, Celery, React, Next.js
Databases & Messaging: PostgreSQL, Redis, Kafka, RabbitMQ
Cloud & DevOps: Docker, Kubernetes, Azure, GitHub Actions
Testing & Observability: Locust, Testcontainers, Micrometer, Prometheus, Grafana, Flower

Education

SASTRA Deemed University 2020 – 2024

B.Tech, Computer Science (AI & Data Science) CGPA: 8.11 | Thanjavur, Tamil Nadu

Achievements

- Solved **1,110+** DSA problems on LeetCode; achieved **Knight** status with a peak rating of 1,854 (top 5%).